

PAPER

Articulation strategies for English liquids used by Japanese speakers

Jeff Moore^{1*}, Jason Shaw², Shigeto Kawahara³, and Takayuki Arai^{1†}

¹ Sophia University
7-1 Kioi-cho, Chiyoda-ku, Tokyo, 102-8554 Japan

² Yale University
7New Haven, CT 06520, USA

³ Keio University
2-15-45, Mita, Minato-ku, Tokyo, 108-8345 Japan

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Abstract: This study examines the articulatory gestures used by Japanese speakers to produce the English liquids /r/ and /l/. Five native Japanese speakers of varying levels of English acquisition are recorded both acoustically and with Electromagnetic Articulography. Seven distinct articulation strategies were identified. Results indicate possible stages of acquisition for a non-native contrast, from a single articulation, through a period of high variation, and finally arriving at distinct gestural strategies for each phoneme.

Keywords: articulatory phonetics, second language acquisition, foreign accent, speech production

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1. INTRODUCTION

1.1 Japanese /r/ and /l/ distinction

English liquids present a unique challenge for the adult Japanese English learner. The Japanese phonological system includes neither an /r/ nor an /l/ phoneme, but has a phoneme which is often considered to be a tap (/ɾ/), distinct from both [1], and frequently used by less advanced Japanese English speakers in place of both English liquids. Learning to perceive and produce these two different sounds is one of the major hurdles a Japanese learner of English faces.

Best's Perceptual Assimilation Model (PAM) [2] argues that learners categorize non-native sounds according to their proximity to their first language, and that differentiating two sounds placed into the same category is among the most difficult tasks for a second language learner. This model has been applied to Japanese speakers' difficulties with /r/-/l/ distinctions[3], with results indicating that Japanese speakers do indeed perceive /r/ and /l/ as a single sound category.

1.2 Variance in /r/ and /l/

The /r/ phoneme differs from /l/ both temporally and spectrally [4] in English, but it can be most readily identified acoustically by a low F3 [e.g. 5, 6, 7]. In order to achieve this acoustic target, speakers employ a wide variety of articulations [e.g. 8, 9, 10, 11]. This variance in articulation strategies extends even to Japanese English speakers, as Masaki *et. al* [12] discovered in an MRI study, finding 4 distinct strategies

for differentiating /r/ and /l/ employed by both native English speakers and Japanese learners.

Other evidence on English liquids suggests that speakers use a variety of articulatory strategies to achieve similar acoustic targets [e.g. 13, 14, 15], producing constrictions at the tongue back or tongue front, adjusting front and back cavities to reach the targeted low F3 typical of an /r/ sound.

1.3 The relationship between perception and production

These findings offer a challenge for the Perceptual Assimilation Model [2], which argues that listeners perceive articulatory gestures directly—that is, the acoustic signal includes articulatory information that listeners hear and interpret directly as gestures. If listeners are truly categorizing sounds according to the perceived articulations, then articulations that differ as wildly as bunched and retroflex /r/ should be categorized separately.

Perception and production of non-native speech sounds appear to be intimately related. Flege's Speech Learning Model [16] asserts that perception precedes production. If a learner cannot learn to perceive two sounds as separate, they will be unable to adjust their own articulation in order to produce the sounds accurately. Flege argues further that no critical period exists after which a sound distinction cannot in principle be learned, and that the mechanisms of first language acquisition remain active throughout life. Although children lose the ability to distinguish non-native sounds after the first few months of life [17], and age of acquisition seems to be one of the most important factors in determining the degree of foreign accent a speaker will eventually retain [18], it is clear that adult speakers do learn new sound distinctions. The

* jeffmoore@eagle.sophia.ac.jp

† arai@sophia.ac.jp

degree to which they become truly native-like remains an empirical question.

It is important not to be too tied to a segmental model of second language perception and production, however. A series of studies conducted by Lively, Bradlow and colleagues [19] followed Japanese English learners as they developed the ability to discriminate between /r/ and /l/. Their results indicate that not all phonetic environments are equally challenging. Initial consonant clusters, such as the /pl/ in “play”, or the /pr/ in “pray”, were found to be particularly troubling. This result indicates that the problem of Japanese /r/-/l/ distinction is not merely a segmental issue, operating at the level of the phoneme, but may be complicated by co-articulatory effects.

1.4 Research questions

Against this theoretical background, this study examines the following questions:

1. How do Japanese speakers articulate English /r/-/l/ distinctions? What variations are present?
2. How do the articulations of intermediate and advanced adult second language learners differ? How are they different from those of a native speaker?
3. To what degree do speakers’ articulation strategies affect the acoustic signal they produce?

2. METHOD

2.1 Participants

Six native speakers of Tokyo Japanese participated. Four participants were students, aged 19 to 23, while two participants were instructors in their late 30s. Participants were balanced on gender, three male and three female, and roughly balanced on level of acquisition, according to a native speaker’s judgment of their accents. Only the two most advanced speakers, both instructors, had spent significant time overseas. Participants were compensated for their time and travel expenses. Procedures were explained in Japanese.

Two male native speakers of English also participated, one from Washington State in the US, the other from Eastern Australia, both in their early 30s.

Participants’ information are summarized in Table 1.

Six Japanese Participants			
(JMA)*	Japanese Male, Advanced	(JFA)	Japanese Female, Advanced
(JMUI)*	Japanese Male, Upper Intermediate	(JFUI)	Japanese Female, Upper Intermediate
(JMLI)	Japanese Male, Lower Intermediate	(JFLI)	Japanese Female, Lower Intermediate
Two Native English Speaking Participants			
(AmMN)	American Male, Native	(AuMN)*	Australian Male, Native

Table 1 Participants

Unfortunately, errors in data collection rendered some samples unusable. Due to head correction issues, JMA’s data could not be included in the current

analysis, while JMUI and AuMN were excluded for extremely erratic positioning data. These participants were removed from the study and are marked with an asterisk in Table 1.

2.2 Utterances

Participants were asked to read sentences from a monitor, presented using ePrime software. They saw a collection of fifteen words embedded in a carrier phrase and presented ten times in random order. The carrier phrase, “Okay, <word>”, ensured that the tongue returned to a mid-level position that is clearly distinct from a Japanese /u/ articulation, which is routinely used for vowel epenthesis to break up non-native clusters (e.g. “play” would be pronounced as /purei/ in the most Japanized pronunciation). The word list included /r/ and /l/ in initial singletons and in initial clusters.

The stimuli in this study are summarized in Table 2.

Set	Members
Initial Singleton /l/ and /r/	laid
	lay
	raid
Initial Clusters with /l/ and /r/	played
	blade
	clay
	prayed
	braid
	crane

Table 2 Stimuli

2.3 Data Collection

Participants had their speech movements recorded using an NDI Wave Electromagnetic Articulograph (EMA), recording at 100 Hz.

Five sensors were affixed to the tongue in order to track the tongue movement for their production of English liquids. Three were affixed along the median sulcus, or sagittal midline. The most anterior sagittal sensor (hereafter TT, Tongue Tip) one centimeter from the tip of the tongue. The most posterior sensor (hereafter TD, Tongue Dorsum) was affixed as far back as was comfortable for the participant, ranging from 5 to 6 centimeters from the tongue tip at furthest extension. A mid-tongue sensor (hereafter TB, Tongue Blade) was affixed at the midpoint between these two. Two lateral sensors were affixed in line with TB, along a line parallel to the sagittal midline. These were placed one centimeter from the right edge of the tongue (hereafter TR, Tongue Right) and one centimeter from the left edge of the tongue (hereafter TL, Tongue Left).

Additional movement tracking sensors were placed at the vermillion border of the upper (UL) and lower lips (LL), as well as on the outside of the lower gums, beneath the lower incisors (JAW). Finally, three sensors were affixed to the nasion (N), between the eyes, and behind the ears at the right (RM) and left (LM) mastoid processes. These last three were necessary in order to correct the data for the movement of the participant’s

head, adjusting all movements in order to plot them in a stationary, three-dimensional space.

Sensor placements are depicted in Figure 1 and summarized in Table 3.



Figure 1 Participant with sensors attached. Photograph used with permission.

Label	Placement
TT	1cm from tip; Tongue Tip
TB	Between TT and TD; Tongue Blade
TD	5-6 cm from tip; Tongue Dorsum
TL	1 cm from left edge; Tongue Left
TR	1 cm from right edge; Tongue Right
JAW	Gum of lower incisor; Jaw
UL	Vermillion border of Upper Lip
LL	Vermillion border of Lower Lip
N	Nasion
RM	Right Mastoid Process
LM	Left Mastoid Process

Table 3 Sensor Placement

Participants' speech was recorded acoustically at 22kHz, using a Schoeps MK 415 supercardioid microphone with Schoeps CMC 6 Ug power module.

2.4 Articulatory Analysis

Participants' articulatory data was head corrected using Donald Derrick's 2012 head correction scripts and analyzed using Mark Tiede's mView package for MATLAB[20].

2.5 Shapes

Certain consistent patterns emerged in the course of analyzing participants' articulation data. Speakers seem to draw from a set of possible strategies in attempting to produce /r/ and /l/. These strategies are described below in terms of the movements of five tongue sensors, and given names for simplicity.



Figure 2, Retroflex: The tongue tip rises, the blade remains fairly steady on the vertical axis, and the whole

tongue body retracts, indicating a pharyngeal gesture at the tongue root. Lighter circles with superscript '1' indicate the sensor's initial position before the gesture, while darker circles with the superscript '2' indicate the sensor position at minimum velocity for the gesture.

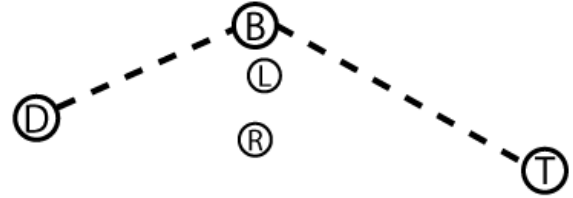


Figure 3, Bunch: The tongue tip remains low, the dorsum rises, and the tongue body retracts. Note also that the left and right lateral sensors are below the blade, indicating a domed shape in the tongue. This strategy appeared most often after /k/ for the intermediate speakers, and was used in all environments by native English speaking participant AmMN.

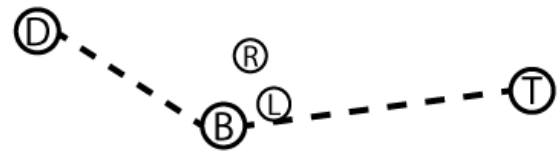


Figure 4, Cup: The tongue tip rises, the mid-sagittal blade sensor drops down between the two lateral sensors, and the tongue does not retract from the resting position horizontally. Although the tongue root was not directly observed in this experiment, the lack of horizontal movement among the five tongue sensors indicates that there was probably no pharyngeal gesture involved.

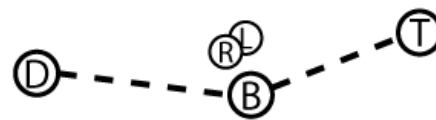


Figure 5, Cup in Retroflex: The tongue tip rises, the mid-sagittal blade sensor drops down between the two lateral sensors, and the tongue retracts. This was the gesture overwhelmingly preferred by Speaker JMLI. Given the speed of the tongue tip in this gesture, JMLI was most likely producing a retroflexed tap, an attempt to adjust the Japanese [ɾ] tapping gesture to achieve a more target-like production.

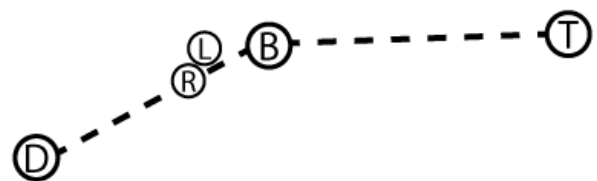


Figure 6, Flat: The blade and tongue tip rise together, while the dorsum remains relatively low.

These gestures probably produce a post-alveolar, lamino-apical closure, typical of Japanese productions of /t/ and /d/ sounds. EMA data does not directly indicate closure, however, so therefore these gestures are labeled ‘flat tongue shapes’ rather than laminar closures.

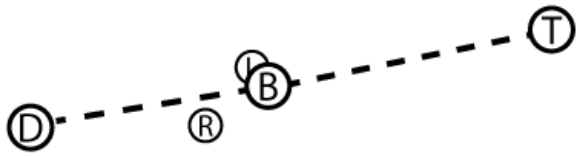


Figure 7, Reach: The tongue tip juts forward, stretching the body of the tongue in a straight line. The blade does not drop, as it does for the ‘cup’ gesture, nor does it rise with the tongue tip. Rather, the tongue tip shoots forward and pulls the blade and dorsum with it. Where the Retroflex gesture is characterized by movement toward the back of the mouth, Reach is characterized by forward motion at the front, a lengthening of the tongue body, and therefore constriction in the front of the oral cavity.

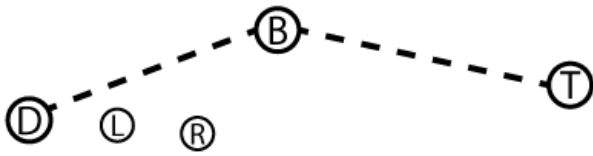


Figure 8, Hunch: The center of the tongue blade takes the lead, rising up above the dorsum, tip, and lateral sensors. These gestures probably indicate a closure at the hard palate. The Hunch gesture is distinct from the Bunch in that the tongue does not retract toward the back of the mouth. The front oral cavity is constricted in a Hunch, while Bunching features a larger anterior cavity and constriction behind the tongue.

2.6 Acoustic Analysis

Acoustic data was analyzed in Praat[21], taking intensity, duration, fundamental frequency, and formant frequencies for each segment. Formant frequencies reported here were taken at the segment’s midpoint.

2.7 Statistical Analysis

Cohen’s d [22] is used here as an estimate of the amount of variance between two groups, the productions of /r/ and /l/. Cohen’s d subtracts the mean of one group from the mean of another and divides by the standard deviation of the set of both groups.

$$d = \frac{M2-M1}{\text{pooled standard deviation}}$$

In this case, a speaker’s /r/ segments constituted one group, and the same speaker’s /l/ segments constituted a second group. Since /r/ and /l/ are distinguished mainly by the third formant frequency, calculating Cohen’s d

for F3 gives an estimate of how effectively speakers differentiated /r/ and /l/.

3. RESULTS AND DISCUSSION

3.1. Variance in Strategies

3.1.1 JFA—Japanese Female, Advanced

JFA has developed a single, distinct strategy for producing /r/ segments, exclusively using the Retroflex gesture. Her /l/s are nearly as consistent, as she uses a Cup for all but two of the tokens. As a highly advanced English speaker, JFA seems to have solidified her production into consistent, reliable strategies, and does not vary them much at all.

Note that there is a single error listed under ‘clay’ in Table 4, but it does not reflect an actual pronunciation error of the speaker. Due to a mistake in setting up ePrime to display the stimuli, one of the ‘clay’ presentations was replaced with the word ‘claim’, and there are thus only 9 total tokens of the word ‘clay’ in this data set. All speakers saw this same presentation, and so all have a single error listed under ‘clay’.

	cup	flat	reach	hunch	cup in retro	retro	bunch	error
blade	10							
played	10							
clay	10							
laid	8			1				1
lay	9			1				
braid						10		
prayed						10		
crane						10		
raid						10		
Total	47	0	0	2	0	40	0	1

Table 4 Articulation Strategies, JFA

3.1.2 JFUI—Japanese Female, Upper Intermediate

JFUI primarily uses the Flat tongue gesture for producing /l/, and primarily uses Retroflex for /r/. She has some variance, however, particularly in her productions of ‘clay’ and ‘crane’. It would appear that JFUI has not yet settled on a single strategy for /l/ or a single strategy for /r/, as our advanced speaker, JFA did. Rather, she is still experimenting to find the right articulation to meet her needs.

She does not use Retroflex or Bunch to produce /l/, and only once does she use Flat to produce an /r/. She seems to have clear-cut categories of the sorts of gestures that she wants to use for an /l/, and the sorts of gestures that she wants to use for an /r/. However, she has not yet settled on particular strategies.

	cup	flat	reach	hunch	cup in retro	retro	bunch	error
blade	1	8	1					
played	1	8	1					
clay	3		5	1				1
lay		8	1	1				
laid		9	1					
braid						8	2	
prayed						7	3	
crane						4	6	
raid		1				7	2	
Total	5	34	9	2	0	26	13	1

Table 5 Articulation Strategies, JFUI

3.1.3 JFLI—Japanese Female, Lower Intermediate

JFLI uses a variety of strategies to attempt to produce the English liquid sounds. In fact, she is the only participant who uses each strategy at least once. Unlike JFUI, she crosses boundaries with her productions, using Retroflex and Bunch in attempting to produce an /l/ after a bilabial plosive. She also uses the Cup strategy seven times in attempting /r/ sounds, as well as the Cup in Retroflex and Hunch strategies once each.

It seems that JFLI also has not settled on a single production strategy for each sound. She has also not clearly defined which strategies are useful for producing which sounds, and mixes the two categories to some degree. She seems to be particularly hard at work on ‘raid’, experimenting with an articulation that will work for her. JFLI has /r/ and /l/ categories, but they are fuzzy and overlap a bit.

	cup	flat	reach	hunch	cup in retro	retro	bunch	error
blade	6	3					1	
played	7	1	1			1		
clay	8			1				1
lay	10							
laid	9			1				
braid	2					8		
prayed	2					7	1	
crane						6	4	
raid	3			1	1	5		
Total	47	4	1	3	1	27	6	1

Table 6 Articulation Strategies, JFLI

3.1.4 JMLI—Japanese Male, Lower Intermediate

Unlike the previous three participants, JMLI does not have preferred strategies for /l/ and /r/. Rather, he has a single strategy, Cup in Retroflex, that he applies to almost three fourths of all tokens. Only ‘clay’ shows a preference for a strategy other than Cup in Retroflex, and there he uses a normal Retroflex.

Recall that these two strategies are distinguished by the relative height of the tongue blade. The Cup in Retroflex is special for having a tongue blade lower than both the dorsum and the tip. The two are acoustically similar, as we will see in the next section.

JMLI does not seem to have developed separate categories for /l/ and /r/ yet. But he is not necessarily applying a Japanese /r/, either. This Cup in Retroflex

strategy is quite different from the sort of tapping gestures typical of Japanese [ɾ], and does have the characteristic drop in the third formant frequency that an English [ɹ] does. He is aware that the Japanese liquid sound in his native language is inadequate to the task, so he attempts an articulation that gets him an acoustic result closer to his target.

	cup	flat	reach	hunch	cup in retro	retro	bunch	error
blade	1				9			
played	2		1		7			
clay		1	1		2	5		1
lay	1	1			6	2		
laid		3	1		6			
braid					10			
prayed	1				9			
crane					8	2		
raid		1			9			
Total	5	6	3	0	66	9	0	1

Table 7 Articulation Strategies, JMLI

3.1.5 AmMN—American Male, Native

AmMN is a Bunch user, applying it to absolutely every /r/ articulation. He does have some variation in his /l/ production, sometimes using a Reach, sometimes a Hunch. Recall that these are fairly distinct movements. A Reach involves the tongue tip stretching up and forward toward the alveolar ridge from a lower, resting position, pulling the rest of the tongue forward. In contrast, the tongue blade leads in the Hunch gesture, most likely making contact a bit behind the alveolar ridge.

AmMN is entirely consistent in his /r/ production, but his /l/ has variation, more even than our advanced Japanese speaker. One possible reason is that JFA, as an adult learner and linguist, has made more conscious effort over the years toward producing good, comprehensible /r/ and /l/ sounds. AmMN, as a native speaker, would not feel any pressure to go beyond ‘good enough’, so he could feel free to vary his articulations as long as they all fall within a certain acoustic threshold.

	cup	flat	reach	hunch	cup in retro	retro	bunch	error
blade			8	2				
played			7	3				
clay	2		6	1				1
lay			6	4				
laid			9	1				
braid							10	
prayed							10	
crane							10	
raid							10	
Total	2	0	36	11	0	0	40	1

Table 8 Articulation Strategies, AmMN

3.2 Acoustics

Participant	Mean /l/ F3	Mean /r/ F3	Mean difference	Cohen's <i>d</i>
JFA	3026	2489	537	2.67
JFUI	2939	2309	630	3.38
JFLI	3125	2748	377	1.48
JMLI	2372	2174	198	0.52
AmMN	3039	2304	735	1.56

Table 9 F3 values in Hz, all participants

Table 9 shows the results of the acoustic analysis. AmMN has the largest mean difference between /l/ and /r/, which should suggest greater perceptual salience in his samples. However, his data had a much wider variance than either JFA or JFUI, with a standard deviation in his /r/ samples of 674.49—nearly as large as the mean difference between categories. Looking at the speakers by Cohen's *d*, we see that JFA and JFUI had much more consistently different /l/ and /r/ productions, although the overall differences between their /l/ and /r/ productions were lower.

JMLI can distinguish the sounds, but he applies the distinction inconsistently. His use of the Cup in Retroflex strategy for nearly three quarters of all utterances was the major contributing factor. Other strategies are more successful, as shown in Table 10:

Gesture	Mean F3
cup	2709
flat	2369
reach	2587
cup in retro	2214
retroflex	2311

Table 10 F3 Values by Gesture, JMLI

JMLI is very capable of producing segments with a low F3, using his Cup in Retroflex and Retroflex strategies, and he can produce segments with high F3 by using either Cup or Reach. JMLI's relatively small difference in F3 between his /r/ and /l/ productions seems to come down to an inability to consistently apply his successful /l/ strategies.

3.3. Acoustic Effect of Articulation Strategies

Generally speaking, participants were able to achieve large distinctions in F3 by using appropriate gestures. Cup, Flat, Reach, and Hunch gestures mostly resulted in segments with F3 values in the high 2000s to low 3000s. Table 11 presents the mean F3 values for each speaker's articulation strategies.

Speaker	cup	flat	reach	hunch	cup in retro	retro	bunch
JFA	3022	3035	—	3108	—	2488	—
JFUI	2980	2942	2917	2853	—	2280	2357
JFLI	3148	3030	3151	2935	2761	2581	3018
JMLI	2753	2249	2587	—	2148	2136	—
AmMN	2825	—	3053	3029	—	—	2304

Table 11 F3 values in Hz by speaker and gesture

JMLI does not achieve quite so successful a distinction with his Flat, and his Reach still only achieves an F3 of 2587, but his Cup gesture is very nearly in the range that other speakers achieved, and his Retroflex and Cup in Retroflex gestures achieve F3 values much lower than other participants'.

JFLI's Bunching strategy does not seem to be very successful in producing an [ɹ]-like sound. In the six times that she employs it, the lowest F3 she ever achieves is 2772, when saying the word 'prayed'. Further, her Retroflex gesture produces a higher F3 than any other speaker's /r/ gestures, suggesting that these two gestures are still works in progress for her.

4. DISCUSSION

4.1 A possible course of acquisition

The articulation strategies employed by these speakers of varying levels gives us some insight into a path by which Japanese English speakers may develop their /r/ and /l/ pronunciations. This study does not track the development of any individual, and a longitudinal will be necessary to make any strong statements on the process of accent development. Nonetheless, a picture does begin to emerge from the data.

In JMLI, our least advanced speaker, we see a second language learner taking the first steps away from his native Japanese tapped /r/ pronunciation. He knows that a Japanese [ɹ] will not achieve English-like pronunciation, so he may be experimenting, mainly with a single gesture that is labeled Cup in Retroflex here. He has not quite divided the /r/ and /l/ sounds into distinct categories yet, and so he has not yet developed different strategies for pronouncing them.

JFLI is moving from an elementary to an intermediate stage, and is developing the distinction between the /l/ and /r/ categories. She mostly focuses on a single successful strategy for producing each sound, but she still experiments extensively. Moreover, she experiments with using articulations that are better suited to the other category. Thus, although she has two categories developing, she is not completely sure which articulations belong to which group. She knows that Cup produces a good /l/, and that Retroflex produces a good /r/, but she is not yet able to consistently apply them. She also has not quite found articulations of either sound that will get her the large contrasts that more advanced speakers achieve, which is probably why she is still experimenting with a few different articulations.

The patterns in JFUI's strategy use are superficially similar to JFLI's strategies, but there is a clear difference. JFUI, as a learner approaching an advanced level, almost never uses an /l/ strategy for an /r/ or an /r/ strategy for an /l/. With the exception of a single token, pronouncing the word 'raid' with a Flat gesture, she uses Cup, Flat, Reach and Hunch for /l/, and uses Retroflex and Bunch for /r/. She strongly prefers the Flat gesture for /l/ sounds, but she's using Retroflex and

Bunch in a 2:1 ratio, indicating that her /l/ is a little more developed than her /r/.

Then at JFA's level, we see a finished articulation strategy. She uses Retroflex exclusively when producing /r/, and is nearly as consistent in using Cup to produce /l/. Through years of effort, she has found two articulations with strongly differing acoustic outputs, allowing her to consistently produce a distinction between /l/ and /r/.

4.2 A possible course of acquisition

We should be cautious, and remember that these are different individuals, and that there is no evidence to suggest that JFA once produced gestures like JMLI, JFLI, or JFUI. But if we extrapolate a little from this limited data set, it appears that Japanese English speakers go through a few stages in developing their /l/-/r/ pronunciation distinction.

They may begin like JMLI, altering the Japanese [ɾ], attempting to lower the third formant frequency and achieve a more English-like sound. At some point, they become more aware of the differences between /r/ and /l/, and like JFLI, experiment with articulatory strategies until they find a few that will suit their purposes. Eventually they begin to understand which strategies are appropriate to each sound, and like JFUI, they restrict their experimenting to only those strategies that fall into a particular sound's category. Finally, they narrow down their articulations to those that are maximally successful, and like JFA, they use one articulation strategy to produce their /l/, and another strategy to produce the /r/ phoneme.

But they do not behave exactly like a native speaker. JFA actually demonstrates less variation in her articulation than native speaker AmMN. This advanced Japanese speaker of English is careful, practiced, and consistent, in a way that the native speaker is not. JFA's strategies are not the same as those used by AmMN. She Cups her /l/; he Reaches and Hunches. She Retroflexes her /r/; he Bunches. She is nonetheless very capable of producing a clear distinction between her /l/ and /r/.

This proposed process for articulatory acquisition very much resembles the classic U-shaped learning model of second language acquisition [e.g. 23,24]. Learners begin with a model of a linguistic structure based in their existing language system, and then appear to increase their error rate as they explore new strategies and expand the input they receive. After a certain point, their error rate begins to decline again, as they narrow down successful strategies and discard unsuccessful strategies.

4.3 Theoretical concerns

The variety of articulatory strategies used here, and the acoustic similarities between them, offer a challenge to the Direct Realist model adopted by Best in her Perceptual Assimilation Model [2]. The speakers in this data set, particularly JFLI and JFUI, appear to be grouping their articulations around an acoustic target, in this case high and low F3. The Direct Realist model

asserts that listeners directly perceive articulatory gestures and categorize them according to their similarity to native gestures. The data here would seem to indicate that speakers are categorizing their gestures in terms of an acoustic target: the freedom with which JFUI varies her articulation of /l/ sounds between Cup, Flat, Reach and Hunch indicates that these gestures are grouped together, according to their acoustic similarities, rather than being treated separately according to their very distinct tongue shapes. Although evidence from a variety of sources supports direct perception of gestures [e.g. 25], this study complicates that picture somewhat, at least in regard to non-native speakers.

5. CONCLUSION

This study provides insight into the strategies Japanese speakers use in articulation English /r/ and /l/ sounds. Our Lower Intermediate speaker uses a single articulation primarily, making no distinction between the two, while experimenting with alternative gestures. The Upper Intermediate speakers have found successful strategies, but they are still experimenting with other gestures and fine-tuning their pronunciation. The Advanced speaker uses a single, consistent strategy for each sound—more consistent, in fact, than the native English speaker, and using different strategies. Her articulation is precise and successful, but not identical to the native speaker's articulation.

In teaching pronunciation, we should be aware of these patterns and understand the process that students take in becoming advanced, non-native speakers. The process starts with developing categories and experimenting to find successful gestures. Gradually a student learns which gestures will work for them, and which will not, and they further learn which gestures are appropriate for which sounds. They may not ever articulate sounds in their second language in precisely the same way a native speaker would, and they may not necessarily need to, as JFA's English is, after all, very good. Rather, an advanced speaker is someone who can make the sound distinctions that the language requires, using the articulatory skills and equipment they have.

We should expect developing students to be inconsistent, to pronounce a sound differently at different times, even in the same linguistic context. If an intermediate student produces a beautiful /l/ on Tuesday, we should not be surprised when her /l/ is less clear on Friday. She may simply be pushing her vocal tract to try new things, experimenting to find her best articulation strategy. And should avoid defining 'advanced' as 'native-like', because an advanced second-language speaker may look very different from a native upon closer inspection.

Future work on this data set will examine the timing of gestures in initial consonant clusters for Japanese English speakers, in the hopes of determining how overlapping of gestures develops. Additionally, we will examine native speakers' judgments of our speakers' utterances to assess how gestural strategies and timing

differences impact comprehensibility perceptions of accentedness. It is hoped that this work will contribute not only to theoretical understanding of the development of foreign accent and its underpinnings, but also provide practical, practicable information for improving second language accent instruction.

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